ECEN 248 - Lab Report

Lab #1

Digital Logic Gates

Section: 524

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Objectives:
The purpose of this lab is to introduce concepts about digital logic gates. As we learn about digital logic gates, we will also learn how to use a breadboard, multimeter, and power supply in the lab. Finally, we will learn how to identify logic gate outputs by testing circuits using the NOT, NAND, NOR, AND, OR, and XOR gates by using the lab equipment (breadboard, power supply…) to construct the gates and test them to verify the truth tables.

Design:
First, we set up the power supply and multimeter for the experiment. While the power supply is turned off, we connected the power supply leads to the breadboard using only channel one (yellow): red cable connected to the power supply's positive terminal and the pin of this cable to the positive power rail on the breadboard, and the black cable connected to the power supply's negative terminal and the pin of this cable to the negative power rail. Then, using another red and black cable, we connected the red one to the top right-most terminal on the multimeter and the black one right underneath that.

For each gate test, we referenced a given paper to determine which chips to place on the board. When inserting the chip, we put it across the center partition of the breadboard and made sure they were placed correctly to avoid shorting the circuits. Also, always make sure the notch on the edge is on the top side when inserting your chip. Then we connected the positive power rail from the DC power supply to the VCC pin and the negative power rail from the power supply to the GND pin of the chip. Next, we had to insert the multimeter pins: we connected the negative (black) pin to the negative power rail (GND) and placed the positive (red) pin on the gate's output pin we wanted to test. All of these placements can be found on the chart that is provided during the lab.

Now, with all the pins in place and the chip placed correctly, we used the wires to put our inputs in. We placed them with the power supply rail if we wanted high and GND for low. Now, we turned on the power supply by pressing the yellow power button over channel one and recorded the voltage output displayed on the multimeter. When reading the voltage, we double-checked if it was in V or mV and made necessary conversions when recording our results. We tested all the input combinations by switching the jumper wires to low and high. This procedure is repeated for all of the logic gates! We made sure that after we finished testing, we turned off the power supply before testing the next input or gate!

Results:
The following tables are the experimental results of testing the digital logic gates.
NOT GATE

A (High/Low)	Y (Volts)	Y (High/Low)
Low	3.988	High
High	0.055347	Low

For the NOT gate, when we test the NOT gate it outputs a HIGH signal for the low input and a LOW signal from the high input. These results are the expected behavior for a NOT gate.

AND GATE

A(H/L)	B(H/L)	AND2 (V)	AND2 (H/L)
L	L	0.111	L
L	H	0.112	L
H	L	0.112	L
H	H	4.421	H

For the AND gate, it outputs a HIGH signal when both inputs are high. For inputs low and low, low and high, and high and low, it outputs a LOW signal. These results are the expected behavior for an AND gate.

OR GATE

A(H/L)	B(H/L)	OR2 (V)	OR2 (H/L)
L	L	0.05596	L
L	H	3.983	H
H	L	3.982	H
H	H	3.982	H

For the OR gate, it outputs a LOW signal when both inputs are low. For inputs low and high, high and low, and high and high, it outputs a HIGH signal. These results are the expected behavior for an OR gate.

NAND GATE

A(H/L)	B(H/L)	NAND2 (V)	NAND2 (H/L)
L	L	3.982	H
L	H	3.978	H
H	L	3.977	H
H	H	0.061857	L

For the NAND gate, it outputs a LOW signal when both inputs are high. For inputs low and low, low and high, high and low, it outputs a HIGH signal. These results are the expected behavior for an NAND gate.

NOR GATE

A(H/L)	B(H/L)	NOR2 (V)	NOR2 (H/L)
L	L	4.001	H
L	H	0.058171	L
H	L	0.058178	L
H	H	0.057409	L

For the NOR gate, it outputs a HIGH signal when both inputs are low. For inputs low and high, high and low, and high and high, it outputs a LOW signal. These results are the expected behavior for an NOR gate.

XOR GATE

A(H/L)	B(H/L)	XOR2 (V)	XOR2 (H/L)
L	L	0.156	L
L	H	4.42	H
H	L	4.42	H
H	H	0.172	L

For the XOR gate, when the input is a low and high or high and low, it outputs a HIGH signal. When the input is a low and low or high and high, then the output is a LOW signal. These results are the expected behavior for an XOR gate.

Conclusion:

We successfully tested and verified the six digital logic gates: NOT, NAND, NOR, AND, OR, and XOR. By experimenting with the breadboard, DC power supply, multimeter, and the gates, we confirmed that each gate's output matched its expected behavior/ truth table when given all the possible inputs (combinations of low and high). At the end of this lab, we have gained experience on how to successfully test and verify digital logic gates using the lab equipment we used throughout the lab.

Post-Lab Deliverables:

1. **What did you like most about the lab assignment and why? What did you like least about it and why?**

What I liked most about this lab assignment was working with the breadboard and figuring out where to place the cables/wires: positive/negative cables from the power supply and multimeter, as well as the wires. The thing I liked least about the lab was how most of my wires in the kit were broken, or a pin would only be on one of the sides, or the wire would fall off in the breadboard.

2. **Were there any sections of the lab manual that were unclear? If so, what was unclear? Do you have any suggestions for improving the clarity**

I think something unclear was where to place the wires for doing the input/output testing, not the power supply and multimeter pins, the actual wires we experimented with, testing the combinations. I think the reason for this is that the lab manual contained a lot of information, and it was hard to process/remember

all of it. Although it was hard to do at first, there wasn't anything to improve clarity because with the help of the TA and going over the manual with others it became easier to understand.

3. **What suggestions do you have to improve the overall lab assignment**

I have no suggestions to improve the overall lab assignment.